

Interactive Mycelium Composites: Material Exploration on Combining Mushroom with Off-the-shelf Electronic Components

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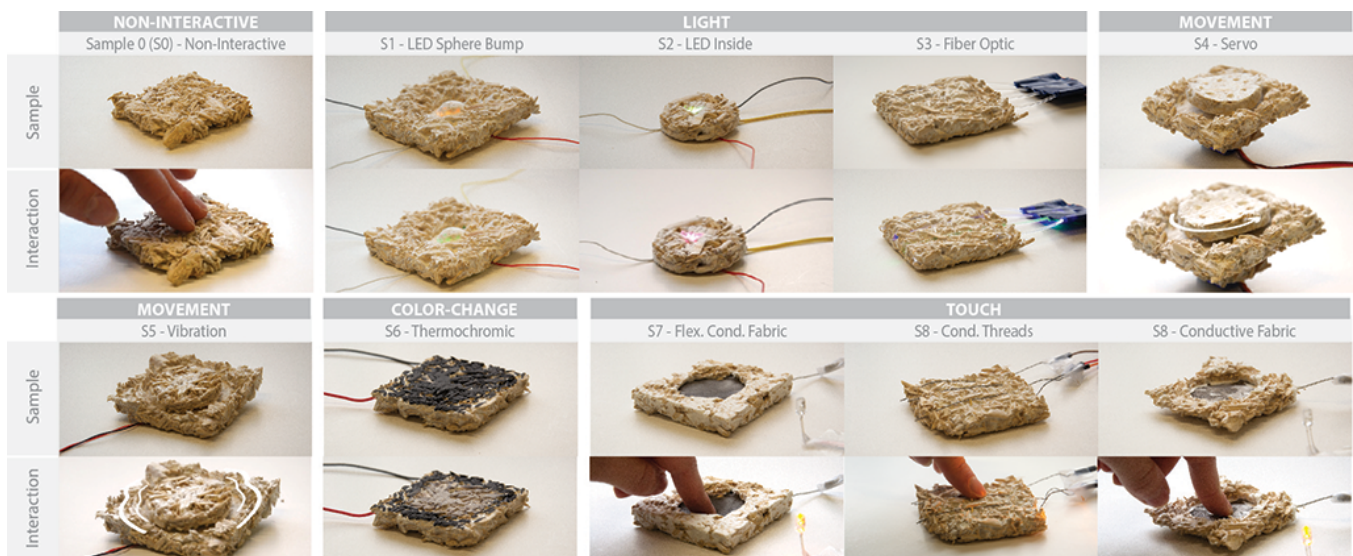


Figure 1: Mycelium samples and interactions organized in categories.

ABSTRACT

Using organic materials for interactive products has gained attention during the past few years. Among them, mycelium, the vegetative part of the mushroom, is a promising material and explored by only distinct examples in HCI. This paper adopts a systematic approach to combining mycelium with LEDs, optical fibers, servo motors, micro-vibration motors, thermochromic inks, conductive threads and fabrics. We created 10 material samples utilizing input (touch) and/or output (light, movement and color-change) features. We describe our material exploration process, as well as the results

of an online survey (N=30) and two design workshops (N=10) which examined the experiences evoked by different interactive elements and collected potential use-cases. Our work contributes insights about the technology inclusion to the natural feel of the mycelium, the material's suitability for tangible interaction and suggests future applications for the composites as a sustainable alternative for interactive interior design elements and ambient displays.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

KEYWORDS

Sustainability, mycelium, material experience, design

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1 INTRODUCTION

Using sustainable materials for the design of interactive products has gained an increasing amount of attention during the past few years. The call for more sustainable solutions is in line with the rising environmental awareness in HCI [5, 30] and among the general public, and linked together with the concerns related to biodiversity loss and climate change. Recycling and biodegradability are viewpoints, which need to now be more strongly taken into account when designing physical artifacts, and material qualities play an increasingly important role. In the spirit of a circular economy and to reduce plastic waste, new pulp, fiber and fungus-based materials are being developed and explored e.g. for the use of the textile industry. Indeed, new organic materials are also appearing in the area of designing interactive technology products, and also HCI has taken steps to explore the new bio-materials through concept design and prototypes.

Mycelium is a mushroom-based material, which has caught the attention of designers and researchers. The material has been offered as a sustainable alternative for many non-interactive applications such as mycelium furniture (i.e. [13]) or wall panels ([23]). Also, explorations with the mycelium exist in HCI but only with distinct examples, i.e. accessories and lighting elements with LEDs [36, 41]. Although they provide insights on how mycelium can be combined with electronics, these studies don't present any evaluations of the experiences that mycelium and technology can yield. On the other hand, material-oriented perspectives highlight the importance of a tinkering process to exploring various alternatives and capturing the potential users' perception of experiential qualities while designing with novel materials [17]. In this respect, a wider and systematic exploration of how to combine electronics and mycelium by also including an evaluation of user experiences of the composite materials is still open for further research. We believe that such an exploration would yield design knowledge on how to expand the application area of mycelium to interactive products. Our aim is to uncover territory and understand the experiential qualities & potential applications of the mycelium and technology combination that we call *interactive mycelium composites*.

To achieve our goal, we first experimented with mycelium and off-the-shelf electronic components for creating mycelium composites that utilize output (light, color-change and movement) and input (touch) features. We implemented 9 interactive mycelium composites (Figur 1), exemplifying three different ways of implementing *light output* with LEDs and optical fibers; two *movement composites* utilizing vibration and rotation; one *color-changing* composite with thermochromic inks; and, finally, *three alternatives* for touch components with conductive threads and fabrics. We presented these samples to participants through an online survey (N=30) and two design workshops. These studies aimed at understanding the material experiences each sample evokes and users' perceptions of potential use cases.

The contribution of our work is twofold: (1) design and implementation of mycelium composites demonstrating light, movement, color-change and touch interactions. (2) the results of the user studies highlight design insights on how these mergers can be perceived by potential users. As our salient findings suggested the organic and natural feel of the mycelium composites are heavily influenced by participants' perception of how well the electronic components and mycelium visually blend. Similarly, the interactive behaviors of the interactive components are expected to complement the organic feel of the mycelium with slow and subtle interactive changes to be perceived as more natural. Finally, the rough and unfinished look of the raw mycelium surfaces felt like the material is uninviting to touch. In light of this information, our work presents insights for future designs of mycelium composites as a sustainable design material.

2 BACKGROUND & RELATED WORK

2.1 Material Oriented Design

The materiality of interaction is a design element that has gained increasingly much attention in designing interactive artifacts [42]. The role of materials in interaction design has emerged since the early days of tangible interaction methods gaining focus in HCI research [12], and now the materiality is emphasized in tangible interfaces (TUIs) [32] and approaches that inherently are linked with physical artifacts, such as internet of things, digital jewelry, and interactive garments. It has been pointed out that focusing on "basic elements of design practice places HCI alongside other materials-centric design disciplines", such as industrial design and fine arts [43]. Indeed, experiencing different materials for computational user interfaces has been investigated with different physical material probes, both as systematic methodological steps [15] and for exploring a certain set of design materials, e.g. samples from nature [11]. Vaaleart et al. have conducted a literature review of 64 studies focusing on the experiential characters of materials in the domain of design and materials [37]. It has also been pointed out, that current tools do not optimally support designers in making material choices, but are quite technical in their approach with respect to how materials are described [18].

In contrast to the technical approaches to designing with materials, the more-than-human design suggests that materials' agency "before, after and during the life cycle of matters" should also be considered [39, p. 160]. In this view, the design is a holistic experience that is shaped by the agency of materials, humans, and structures that tie counterparts to each other. For example, the experience of Tilting Bowl [40], a ceramic bowl that randomly and subtly tilts, includes the agency of clay, water and heat interacting with each other in certain ways, modifying the shape of the bowl, while in production and also in use. Furthermore, with the inclusion of technology, i.e., servo motors and PCB controllers, this artifact becomes a computational one that includes the features of randomness and temporality, shaping the experience together with human interpretation. Similar perspectives are also presented to examine the agency of autonomous data technologies, such as Artificial Intelligence and Internet of Things [9] that these technologies shape reality by autonomously interacting with other counterparts (i.e., creating playlists for humans, exchanging data with each other).

Overall, more-than-human design highlight the need of examining the lived experiences of matters by paying attention to specific ways that the counterparts, including the materials, shape the reality with their agency, rather than focusing on unidirectional paradigms that consider the human as the central part of the experience.

In terms of examining the experiences with materials thoroughly, Giaccardi and Karana describe the framework of the material experience [10]. They suggest that materials are experienced through four experiential levels: sensorial, interpretive, affective, and performative, and how this framework can be used to analyze material experience patterns. On the sensorial level, materials are encountered through the human sensory system (touch, vision, smell, sound and taste). On the interpretive level, the material is judged by the meanings people give them. On the affective level, the emotions evoked by the material are examined, for example, the disappointment or surprise caused by the material. The performative level describes how the user responds to the material. Karana et al. propose the Material Driven Design Method for use in projects where the design is based on material and user experience [16]. The method includes four steps: (1) Understanding the material: technical and experiential characterization, (2) Creating materials experience vision, (3) Manifesting material experience patterns and (4) Designing material/product concepts. Every step of the method is not always necessary, and the steps can be chosen to fit the goal of the overall design project. Also, Vallgård and Redström propose an approach to considering both digital and physical affordances of the computational artifacts, in other words, computational composites [34]. Vallgård argues computers as a design material that provide interactive features like temporality, connectivity and reversibility, as well as the physical qualities of conventional materials that they are combined with [33].

Overall, the material-oriented design approaches suggest that the design exploration of novel interactive materials should consider critical steps of tinkering with materials by focusing on interactive, as well as physical, opportunities and understanding experiential qualities of the composites.

2.2 Designing with Mycelium

Emerging bio-degradable materials are currently being introduced in material explorations for designing interactive products, mycelium among them. Mycelium is the vegetative part of a fungus, and mycelium materials consist of mycelium and the growth medium. Mycelium requires nutrition and the right conditions, including humidity and heat, for growth. Mycelium acts as a binder between the other materials. The properties of the material vary depending on the composition of the medium and the selected fungal variety. Waste produced by, for example, agriculture or construction industries can be used as a growing medium. Properties and manufacturing methods of the mycelium are well studied in prior work. Designing with mycelium-based materials using the MDD method has been explored [17, 27], and Parisi et al. presented mycelium as a material for experience design due to its dynamicity and spontaneity [27]. Sauerwein has studied the aesthetic appreciation for mycelium-based materials [29], and Arias Barrantes [2] explored leather-like mycelium materials. Mycelium materials also already have some commercial applications. For example, Ecovative Design

makes packaging from mycelium and has grown interior design products such as lampshades. MOGU manufactures acoustic panels from mycelium. MycoWorks, Mycotech lab and Mylo produce substitute materials for animal leather for the use of the clothing industry.

The prior studies have also explored mycelium and interactive electronics. Weiler et al. [41] explored biofabrication of Mycelium artifacts, including electronics (i.e., LEDs and LCD displays) embedded in the growing process. Vasquez and Vega explored prototyping DIY electronics with Mycelium [35] and presented the ways electronic components can be embedded into pre-produced mycelium sheets. Vasquez et al. present a Life Cycle Assessment approach applied to the topic, and demonstrate a sustainable life cycle for wearable electronics with mycelium [36]. In addition, a workshop with novice designers and the positive impact of using mycelium boards as an interactive prototyping material are highlighted [20]. These studies have presented quite distinct examples of combining electronics with mycelium. In this paper, we seek to approach the topic systematically, experimenting with a wider set of electronic components to create composite materials, and comparing the experiences evoked by different interactive elements.

3 MYCELIUM EXPERIMENTS

We created material composites with electronic components and mycelium. Our aim was to explore composites that can function as *input (touch)* and *output (light, movement and color-change)* modalities for interactive applications. Our general tendency was to use off-the-shelf components, that are commonly used in interactive prototypes, in combination with the mycelium. In each category, we identified the components to combine and sketched out alternatives for the composite samples to experiment with. **Figure 1** summarises the samples. Here, we should note that our selection criteria for the samples and the categories were framed by the common usages of the off-the-shelf components in interactive artifacts. For instance, light and the thermochromic color change is commonly used to visually output information in interactive devices [14, 28], whereas the thermochromic one could have been framed with its input functionality, changing color with the human body temperature [19]. Or the movement sample with rotating pieces could have been used as an input [24]. Yet, in our study, we selected and framed the samples with the primary functions of the technology used in the samples for gaining more focused feedback from the participants.

For all composites, we used a commercially available mycelium growing kit (Grow-it-yourself kit from Grown.bio). We selected this kit, with relatively large wood chips as a growth medium, due to its ease of availability for others to repeat the growth process. All the samples were grown according to the growing instructions of the kit ¹ and the two-stage-growing process advised by [36], see Figure 2. Based on this information, in the first growth phase, we activated the mycelium by mixing the advised amount of flour and water with the growing medium inside its bag. We covered the bag with stretch film and poked holes with a sterilized knife to let mycelium breathe. We stored the bags in a dark environment at room temperature (20–25°C) for 8 days. At the end of this phase,

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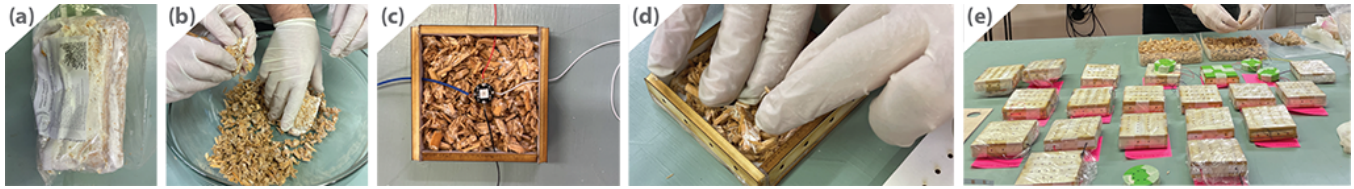


Figure 2: An overview of the growing process: (a) Mycelium bag at the end of the first growing phase; (b) crumbled mycelium; (c) the mold of Sample 2 with the component; (d) mycelium medium and component in the mold; (e) grown sample.

the mycelium started to grow, covering the medium with its white-colored structure. For the second growth phase, we crumbled the mycelium into little pieces and put them into our custom-designed molds to start the second growth phase.

Almost all molds were designed to produce 8x8 cm samples with approximately 1 cm thickness that would provide enough volume to put selected electronic components. While these dimensions provided a general guide for us, we interpreted these for individual samples with larger components, i.e. servo motors. The molds were produced with Polystyrene foam sheets and complemented with 3D printed pieces where complex geometries were needed. The foam sheets, as well as the 3D printed parts, had small holes for the mycelium to breathe while growing. Specific considerations while preparing the molds of different samples are mentioned below within the sample descriptions and illustrated in Figure 3. The mycelium crumbs put into the molds are lightly pressed, leaving room for the mycelium to grow. Before including the electronic components in the molds, we soldered thin cables to the components and briefly sterilized them with a paper towel and alcohol. The soldered cables were long enough to extend them out of the molds, enabling us to utilize the functionality of components after the growing phase. After putting the medium and the electronics into the mold, they are closed tightly by using tapes between mold pieces to ensure the closure. The second growing phase lasted 7 days after which we demolded the samples. After demolding, all samples were connected to Arduino UNO. Now, we describe the design of samples in each category:

3.1 Light Samples

In the light composites category, our purpose was to come up with structures where a computable light source is placed in the mycelium composites, allowing us to create light effects on the surface. For this aim, we chose RGB LEDs circuits (Flora RGB Smart NeoPixel version 2) and side-glow optical fiber ($\varnothing 1.5\text{mm}$) with 3mm through-hole LEDs attached.

In *Sample 1 (LED sphere bump sample)*, our aim was to leverage the capacity of mycelium growing in 3D shapes and to explore light cast through the material. To explore this, we created an 8x8 cm mold with a thickness of 1 cm. The mold included 3D printed parts in the center; a male sphere on the bottom and a female sphere on the top, leaving a gap of 2 mm for the mycelium to grow in (Figure 3b). We then filled the mold with the growing medium and closed the mold. As a result, the grown block included a sphere-shaped bump in the center. To cast the light, an RGB LED circuit was placed under the sphere part of the block. In *Sample 2 (LED inside)*, we explored growing mycelium over the RGB LED circuit.

For this sample, we experimented with a smaller cylindrical form factor ($\varnothing 45\text{mm}$) for the mycelium blocks. We created a cylindrical mold with 3D printed parts (Figure 3c) and we put the RGB LED circuit directly inside the growing medium on the outer surface with mycelium covering the circuit, leaving only the LED visible. The RGB LED circuit had wires soldered and extended out of the mold. For demonstrating these samples, we connected them to Arduino UNO and used Adafruit's Neopixel library to program LEDs to demonstrate a rainbow effect that cycles over time.

In *Sample 3 (optical fiber)*, four optical fibers were placed on the outer surface of the molds (Figure 3d). The mycelium medium covered some parts of fibers to fix them on the mycelium block during the growing phase as we aimed at exposing parts of the wires on the surface. We attached LEDs at the end of the fibers after the sample is grown. The demonstration of the sample was achieved by programming LEDs to fade in and out in appr. 6 seconds intervals.

3.2 Movement Samples

We selected a servo motor (SG90) and a micro-vibration motor circuit (LilyPad Vibe Board) to explore movement on the composites. The composite with servo motor (*sample 4 - servo*) includes two blocks grown in separate molds (Figure 3e): A base block and a piece part. To create the base block, we buried the servo motor in mycelium, leaving only its output shaft uncovered. For the piece part, we created a 3D printed mold that has a standard servo arm placed on the bottom. The servo arm is placed in its mold in a way that the gear of the arm is not covered by mycelium. This approach enabled us to attach two parts together after demolding and program the piece part to rotate. *Sample 5 (vibration)* utilizes the micro-vibration motor inside the sample and also has two pieces (Figure 3f): The base block is made out of mycelium only and has a cylindrical dent on its center. The piece part is a cylinder mycelium block which consisted of a micro-vibration motor circuit placed inside the piece. To assemble these two parts, the cables of the circuit are poked through two holes on the base piece, enabling us to program the piece part to vibrate, creating micro-movements. To demonstrate the interaction of *servo sample*, we coded the servo motor to turn slowly back and forth.

3.3 Color-change Sample

For color-changing composites, we focused on small Peltier modules (40 x 40 x 4 mm) and thermochromic inks. *Sample 6 (thermochromic)* was created with the Peltier module put inside a square mycelium block (Figure 3g). For this sample, we applied thermochromic screen printing ink (SFXC, which switches from black to a transparent state over 27°C) onto the mycelium block after its growth. For

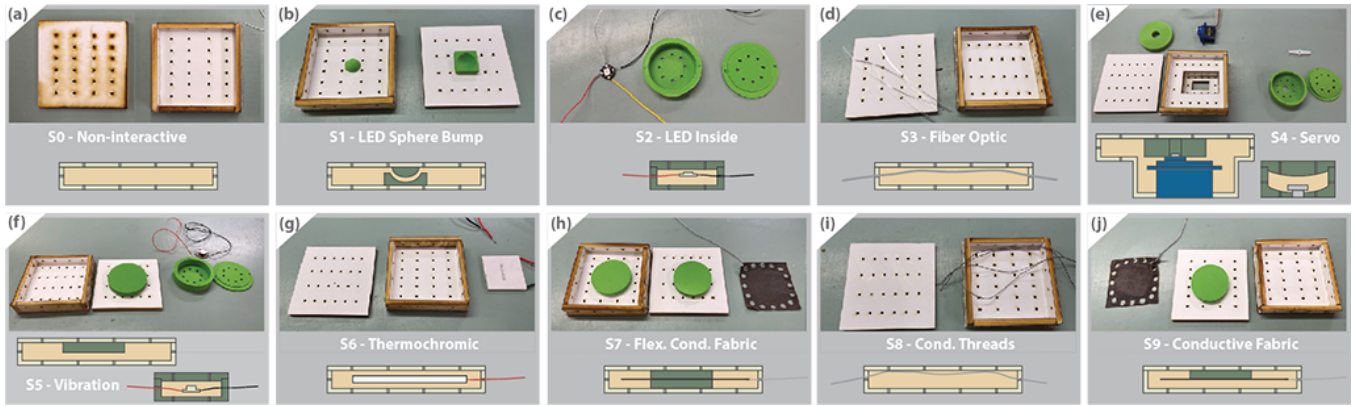


Figure 3: Photos (top rows) and cross sections (bottom rows) of molds: (a) Polystyrene mold; (b) Polystyrene mold with 3D printed sphere bump parts; (c) 3D printed cylinder mold with a LED; (d) Polystyrene mold with fiber optic wires; (e) Polystyrene mold with a servo motor and 3D printed mold with a servo arm; (f) Polystyrene mold with a 3D printed cylinder and 3D printed mold with a vibration motor; (g) Polystyrene mold with a peltier module; (h) Polystyrene mold with 3D printed cylinders and a conductive fabric; (i) Polystyrene mold with conductive threads; (j) Polystyrene mold with a 3D printed cylinder and a conductive fabric.

demonstration, we connected this composite to a bench power supply and applied approximately 8 volts until the color change occurs.

3.4 Touch Samples

The touch composites were achieved by using conductive threads and fabrics. Three samples were designed: In *sample 8 (cond. threads)*, we placed four conductive thread pieces on the outer surface of the mold with mycelium medium (Figure 3i). We covered only some parts of the threads with the mycelium medium to let mycelium grow over the threads, fixing them in place. Similar to optical fibers, we kept threads longer to connect them to a microprocessor for demonstrating their touch functionality. *Sample 7 (flex. cond. fabric)* and *9 (conductive fabric)* incorporate stretchable conductive textile pieces for detecting touch. For both samples, we first cut holes on the edges of the textile pieces and then we stitched a conductive thread on one corner of the pieces. The mold of S9 included two cylindrical 3D printed pieces on the bottom and the top part of the mold (Figure 3h). The bottom half of the mold was filled with mycelium medium until the height of the first cylinder piece. After this, the fabric piece with the conductive thread extension was placed in the center. In this stage, we placed the extension thread over the mycelium medium and let it go through one of the small holes in the mold. The conductive thread extension was used to detect touch. Over the fabric piece, we put the second cylinder aligned with the one on the bottom half and filled the empty spaces with mycelium medium. While demolding, we removed the cylinder pieces and left with a composite where the textile is fixed in the middle and exposed on both sides of the block.

The mold of the *flex. cond. fabric* (Figure 3j) was build in a similar fashion. However, on this composite, we omitted the cylinder piece on the bottom part. This yielded a block where the textile is exposed from only one side. The functionalities of all touch composites were

demonstrated by connecting them to Arduino UNO together with an additional LED lighting up as feedback when they are touched.

3.5 Non-interactive Mycelium Sample

We also grew 8x8 cm mycelium blocks without adding any components or interactive features to them (Sample 0 - *non-interactive sample*). We aimed at using them in the study as a sample to understand the perception of differences when a mycelium block without technology added is compared to other samples.

4 USER STUDIES

In the user studies, we were interested in understanding experiential qualities [18] and potential use cases of the interactive mycelium composites. To achieve these results we conducted an online survey and two design workshops.

4.1 Material Experience Survey

4.1.1 Procedure. Online surveys are a widely used method in HCI, and their use in exploring people's perceptions of novel concepts presented with videos has earlier been used e.g. with on-skin color-changing interfaces [45], telepresence robots [31], or drone-human interaction [3]. Especially, due Covid-19 pandemic and limitations to [6], video-enhanced online surveys provide a feasible opportunity still to conduct user studies. The online survey started with a consent and background information form. After this, the survey continued with the introduction of the categories, explaining to participants that they will be shown videos of samples in five categories: (1) mycelium, (2) mycelium & light, (3) mycelium & movement, (4) mycelium & color-change, (5) mycelium & touch.

Upon proceeding, in each category, we asked participants to watch the video of the sample and answer three questions regarding that sample. The videos demonstrated the interaction of the samples (see Figure 1) from a perspective and top views. In light, movement, and color-change categories, the first question was to

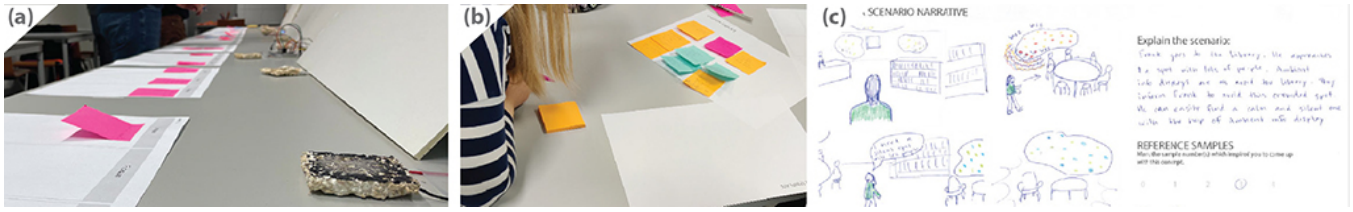


Figure 4: Pictures from the workshops: (a) participants exploring the samples; (b) brainstorming session; (c) sketch of the mycelium-based ambient display.

rate with a 5-point Likert scale (Strongly disagree - Strongly agree) their perception of whether “the change is visible” and “the change is aesthetically pleasing” on the samples. The second question, in all categories, was to select at least 3 meanings out of 20 options that describe their associations with the sample. The third question was to think about which emotions the material elicits from the participants and select at least 3 emotions from the vocabulary provided (20 options). These vocabularies were adapted from Camera and Karana’s toolkit for the experiential characterization of materials [18]. At the end of each category, we showed all the samples of the category and asked two open-ended questions: The first question was about the most pleasant quality of the composites whereas the second question was focused on the most disturbing qualities. We asked participants to mention the sample number if they had specific comments about it. All categories were presented in random order. This randomization was also applied to the samples presented inside the categories.

After all of the categories are presented, we asked an open-ended question about the most unique quality of the “mycelium composites”. After, participants were directed to describe at least one future product application. While answering these questions, participants had the opportunity to watch all of the sample videos shown in the survey. We asked that they indicate the sample number (i.e. Sample 1) if they had specific comments about it. The survey took approximately half an hour for participants to fill.

4.1.2 Participants. 30 participants (18 female and one preferred not to say) responded to the survey. Participants were between 19-47 years old. Five of the participants were knowledgeable about mushroom-based materials and none of them had worked with mushroom-based materials before. Participants attended the survey from four different countries (anonymized). The quantitative responses of one participant were lost due to a software error, therefore, we only included their qualitative responses in the analysis. All the participants were reached through social media calls and individual networks of the authors.

4.2 Design Workshops

4.2.1 Procedure. Design workshops are a well-known method used in exploring different design-orientated questions, and are a good approach when material-centric concepts and tangible prototypes are considered [4, 7, 26, 35]. Our workshop (see Figure 4) started with participants filling out a research permit and a background information form, then, we introduced a presentation of the schedule and the topic. After this introduction phase, as a first task, we

requested participants to explore the samples and use post-it notes to highlight (1) the unique, (2) the most pleasing, and (3) disturbing qualities of the samples [18]. During the task, each sample was connected to Arduino Uno’s and demonstrated their interactivity. The Arduinos were hidden behind white boards to prevent the potential influence of the micro-processors on the responses (Figure 4). During this task, we encourage all participants to interact with each sample. After the first task, we briefly discussed their experiences with the samples.

In the second task, participants were asked to brainstorm concept ideas based on their impression of the Interactive Mycelium Composite samples in small groups (in pairs or in groups of three). They used a template to present their final ideas, see Figure 4. These templates include empty space for concept name, use case, user, use, need, sustainability, sketching scenario narrative, explaining the scenario and referred samples. After the ideation session, each group presented their final concept. The presentations were followed by a joint discussion. Presentations and discussions were video-recorded for later analysis. Each workshop lasted approximately two hours.

4.2.2 Participants. 10 participants attended the workshops (5 per workshop). Half of the participants were women. Participants were students ($N = 7$), researchers ($N = 1$), research assistants ($N = 1$), entrepreneur (photographer) ($N = 1$) aged between 23-42. All the students have backgrounds in arts and design, textile and industrial design. The researchers and research assistants were working in the industrial design faculty, specializing in interaction design. One research assistant and one student had previous knowledge about mushroom-based materials before the workshop and none of them had worked with mushroom-based materials before.

5 FINDINGS

5.1 Aesthetics and Visibility of Change

Figure 5 summarises the mean responses to the visibility and aesthetics of the changes. In terms of visibility, the participants ranked the changes of the *servo* ($M=4.47$), the *LED inside* ($M=4.43$) and the *thermochromic* ($M=4.4$) as the most visible ones. The *vibration* ($M=4.28$) was the fourth most visible composite. On the other hand, the least visible changes were marked as the *LED sphere bump* ($M=3.93$) and the *optical fiber* ($M=3.93$). Mean ratings are all above the neutral threshold, meaning all samples are visible to the participants. To formally compare the visibility of change on all samples, we conducted a one-way ANOVA. This test is used to compare independent groups (the samples) to identify whether there is a

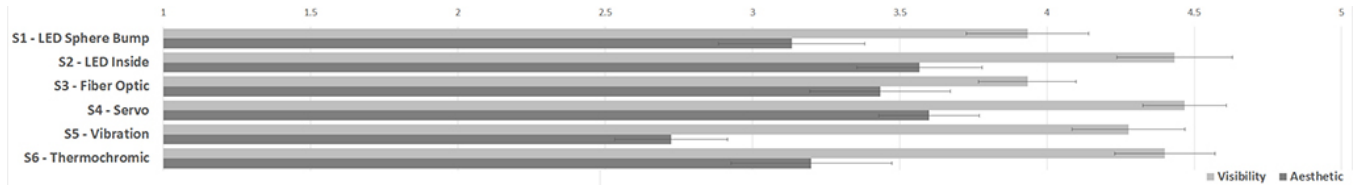


Figure 5: Mean responses to questions about visibility and aesthetics

statistically significant difference among means of the responses (visibility of change) to the independent groups. We did not find any statistically significant effect ($F(5, 173) = [1.881]$, $p = 0.10$).

The mean responses highlighted the *servo* ($M=3.6$) and the *LED Inside* ($M=3.57$) as the most aesthetically pleasing samples. These are followed by the *optical fiber* ($M=3.43$), the *thermochromic* ($M=3.2$), and the *LED sphere bump* ($M=3.13$). The least aesthetically pleasing sample is the *vibration* ($M=2.72$) which is slightly below the neutral threshold. For this parameter, an one-way ANOVA did not showed any statistical significance ($F(5, 173) = [1.881]$, $p = 0.10$).

5.2 Material Experiences

In this section, we synthesize and report the findings from both quantitative responses and qualitative feedback from the survey and the post-it notes of the workshops. The analysis of qualitative responses is achieved through qualitative coding [22]: We first combined the data from both studies. Two of the authors familiarized themselves with the data. Then, they coded the data separately and meet to discuss and reach an agreement several times. The relevant quantitative responses towards emotions and meanings associated with individual mycelium composites are used to support the themes that emerged from the feedback. Figure 6 presents the top four meanings and emotions associated with each sample. While presenting the findings, we refer to individuals who filled the survey as survey participants (SP). As the data for the workshop consisted of post-it notes without any indicator, we use an abbreviation of the post-it note with the indication of which workshop it was mentioned, i.e., PNW1.

5.2.1 Visual integration of the interactive components to the mycelium.

Our results highlight that the visual integration of interactive components on samples is an important factor, complementing the natural feel of Mycelium. In this regard, *non-interactive sample* is associated with *natural* by most of the survey attendees ($N=24/29$, Figure 6). The qualitative feedback pointed to the natural appearance of the *non-interactive sample* as a pleasant aspect (SP4, SP19, SP20, SP24, SP25, SP26, PNW1). The rugged look of the mycelium and its texture are mentioned as contributing factors to this sample's natural feel (SP4, SP5, SP17, SP29): "it has a texture, good to feel when compare a plain plastic or metal surface" (SP17). In this regard, some participants (SNW2, SP15, SNW1, SP9) found it pleasing that the sample resembles other natural materials such as cork (SP15, PNW1), and sunflower seeds (SP6), and something edible like a rice cake (PNW2).

In addition to this natural perception of mycelium material, our samples (*optical fibers*, *flex. cond. fabric*, *conductive threads* and *conductive fabric*) presented different design approaches to integrate

interactive components as visual elements in the composites. In the following paragraphs, we report participants' responses to our samples:

The *Cond. Threads sample* with threads blending on the surface was found well-integrated and makes threads perceived as a part of the whole composite (SP24, SP8, SP27 and SP29). Here, SP8 pointed out the contribution of "the implicit usage of the threads making it more complex and attractive". On the other hand, the participants criticized the visual integration of the conductive textiles on the *Flex. Cond. Fabric* and the *Conductive Fabric* samples, as the textile felt like an addition rather than being a well-blended visual part in the overall design (SP1, SP6, SP20, SP21, SP24): "Flex. Cond. Fabric and Conductive Fabric samples feel a bit eclectic. They just frame the touching area without having any relationship with the interaction" (SP24). SP6 further elaborated that "It looks like we could use any material with a hole inside". Despite the criticism, in the survey, one of the top meanings associated with *Conductive Fabric sample* was the item *natural* (Figure 6). Here, post-its from workshops ($N=3$) emphasized the mycelium fibers growing on the conductive fabric. According to these, the blending of the mycelium and fabric demonstrated a pleasant "combination of two different styles/texture/effect" and evoked a feeling of "nature get/take power over human-made things" (PNW1).

Similar critics were apparent in the responses to *optical fiber sample*: Whilst participants liked the smooth and less noticeable lights within this mycelium block (PNW1, 2xPNW2), the visible optical fibers were noted to be disturbing to the aesthetics of mycelium: "We see too much of the fiber" (PNW1). "Transparent light tubes are not aesthetically pleasing on top of mycelium" (SP19). SP27 reasoned this situation by saying "The light was too artificial looking". On the other hand, two post-it notes mentioned the contrast between optical fibers and the organic look of the mycelium is pleasing. Aligned with the latter comments, one of the top-rated meanings associated with *optical fiber sample* was *futuristic*.

5.2.2 Interactive behaviors of the components in the mycelium composites.

Another apparent theme in responses was to interactive behaviors of the electronic components within the mycelium composites. Participants' feedback suggested that the organic feel of the mycelium should be considered when designing temporal behaviors.

Overall, the qualitative responses highlight that slow movement is more complementary and suitable for mycelium. Here, the most notable example was the *thermochromic sample*. In comparison to other visible changes on composites (i.e. movement and light), the transition of color in the *thermochromic sample* was relatively

S0 Non-interactive (N=18) *Curiosity (N=9) *Comfort (N=8) Fascination (N=8) Confusion (N=6) Doubt Natural* (N=24) H.Crafted (N=16) Strange (N=11) Calm* (N=10) Cozy* (N=10)	S1 LED Sphere Bump (N=19) *Curiosity (N=11) Doubt (N=8) Confusion (N=7) Disappoint. Strange (N=16) Calm* (N=11) H.Crafted (N=10) Futuristic (N=8)	S2 LED Inside (N=9) Curiosity (N=8) Attraction (N=8) Confusion (N=8) Amusement (N=7) Surprise (N=6) Fascination (N=6) Comfort (N=6) *Melancholy Strange (N=11) Toy-like* (N=10) Futuristic (N=9) Calm* (N=8)	S3 Optical Fiber (N=10) Curiosity (N=9) Doubt (N=8) Attraction (N=8) Amusement (N=8) Boredom (N=7) Reluctance Calm* (N=14) Natural** (N=11) Futuristic* (N=10) H.Crafted (N=10) Strange (N=8) Elegant (N=8)	S4 Servo (N=16) Curiosity (N=11) *Amusement (N=10) Confusion (N=9) *Comfort Calm* (N=14) Toy-like* (N=13) Manufactured (N=9) H.Crafted (N=7) Cozy (N=7) Natural** (N=7) Ordinary* (N=7)
S5 Vibration (N=14) Confusion (N=11) *Distrust (N=8) Doubt (N=8) *Rejection (N=7) **Curiosity (N=7) Surprise (N=7) Reluctance (N=7) *Frustration Strange* (N=17) Aggressive* (N=16) Toy-like* (N=10) Not Sexy* (N=10) Vulgar* (N=10) H.Crafted (N=7)	S6 Thermochromic (N=16) Curiosity (N=13) *Fascinati. (N=12) Surprise (N=11) Confusion Strange* (N=17) Natural* (N=15) H.Crafted (N=9) Futuristic (N=9) Not sexy (N=7)	S7 Flex. Cond. Fabric (N=14) Curiosity (N=12) *Amuse- (N=11) *Comfort (N=7) Attraction Calm* (N=12) H.Crafted (N=11) Strange (N=10) Cozy* (N=10) Natural** (N=9)	S8 Cond. Threads (N=17) Curiosity (N=11) *Fascina- (N=8) Confusion (N=7) Doubt Strange (N=15) H.Crafted (N=12) Toy-like* (N=11) Natural** (N=9)	S9 Conductive Fabric (N=14) Curiosity (N=10) Doubt (N=9) Fascination (N=7) Boredom (N=7) Rejection H.Crafted (N=12) Natural* (N=12) Strange (N=9) Calm* (N=8) Ordinary* (N=8) Manufactured (N=7)

Figure 6: Frequency of the top four meanings and emotions for each sample: Left columns show the emotions and the right columns include meanings.

slower. This slow and smooth transition was appreciated by the participants (SP1, SP17, SP25, PNW2): i.e., *"the change is slow and subtle"* (SP25), *"unobtrusive and smooth transformation is fun"* (PNW2) and *"magical"*. SP2, SP20, SP21 and a PNW2 further highlighted the transition's natural and organic manner as a positive aspect. Supporting these participants, SP1 stated *"The slowness of the change and the natural colors, which fits with the mycelium natural aspect"*. On the other hand, the change speed was a disturbing issue for the observability of the *thermochromic sample*: *"The change starts pretty slow so it's hard to see it at the beginning"* (SP30). *"One needs to focus deeply to observe the change. This is a bit tiresome"* (SP21). Another example highlighted in the responses was the *servo sample*. The movement of the mycelium in this sample was also well-received and specifically mentioned by the participants (N=10). They were pleased with the slow and smooth rotation on the composite: *"[Servo sample] had a nice, steady and slow rotation speed, which was nice to watch."* (SP28). SP18 mentioned a reason why the slowness of this material is more relevant to mycelium composites: *"[Servo sample] feels in line with the material's quality. It is mundane, it is hypnotizing like watching a clock. Given that the material is also growing slowly, I found the motion reflecting that aspect."* One participant also mentioned the fusion between mechanical movement and organic look as a pleasing factor (PNW2).

On the other hand, light composites and the *vibration* composite were criticized by participants referring to the aggressive changes. Regarding the *LED inside sample*, the post-it notes pointed out that the light *"moves and blinks too fast"*. SP30 mentioned light is disturbing *"if the light is too strong of a contrast by itself"*. Similarly, 14 survey respondents and two post-it notes mentioned the movement of the *vibration sample* as a disturbing factor: *"Shaking/vibration motion is very big"* (PNW2) and *"The aggressive vibrations make it seem like there is an angry animal inside or it's ready to explode."* (SP29). Quantitative responses also reflected this uncanny feeling that this sample was marked with negative emotions such as distrust, rejection and frustration, as well as meanings like aggressive, not sexy and vulgar (Figure 6).

Apart from the speed of change, participants also commented on the form of change in relevance to the organic feel of the mycelium. In this regard, SP21 mentioned *"The circular geometry of the dynamic part in both movement samples made me question the relation of the material and the movement. If this is such a sloppy-looking material, maybe the movement can also be irregular."*, suggesting irregular movement could reflect the unfinished feel of the mycelium. One example of this irregular movement was highlighted by referring to *thermochromic sample*. In this sample, the non-flat surface of the mycelium led some parts of thermochromic ink on the surface, that is closer to the heat source, warming up faster and, therefore, yielding quicker color changes. Referring to this, SP21 mentioned *"the form of the change is organic. It is not geometrical"*. Also, SP9 found it pleasant that *"Color change of the surface and unevenness reminds me of the worn wall surface of an old building"*. SP4 found the patterns on the surface resemble Jackson Pollock paintings. These paintings usually demonstrate pouring and dripping techniques where the effect looks almost accidental. Supporting these comments, *natural* item was selected as one of the most associated meanings for this sample (Figure 6).

Finally, the sound was also an issue mentioned in the movement samples. One post-it note suggested the buzzing sound fits the *vibration* composite's structure. On the other hand, another note reported that *"organic materials and insect-like buzzing is rather creepy"* (PNW2). Four other post-its found the sound somehow disturbing in the *Servo sample*: i.e., *"Sound don't go well with the organic look..."* (PNW2).

5.2.3 Interaction Affordance for Touch in the Mycelium composites. Another theme that emerged for touch composites was about *the affordances for touch interaction*. In this regard, the survey respondents who observed interactions that we presented in the videos, commented that the *cond. thread sample* can be more pleasant and meaningful for touch interaction as the user would touch on the mycelium surface directly (SP6, SP20 and SP21): For instance, SP21 stated *"[Cond. threads sample] is the best in my opinion. I think what I like most about it is, that the interactive surfaces are embedded in the*

material. So to interact with it, one actually needs to touch mycelium. I think this is a nice quality." Agreeing with this opinion, SP6 and SP20 also wished that the conductive threads were invisible, thinking that it would augment the physical interaction with mycelium texture (SP6) and make the interaction more sophisticated (SP20). Also, SP24 highlighted playful interaction opportunities in multiple conductive threads, looking like a guitar on *cond. thread sample*.

However, the underdeveloped look of the touch samples prevented the user from feeling like they can touch the material. SP26 stated "*the material looks unfinished, it would make me unsure whether I can touch it*" and SP17 was disturbed by the rough surfaces of touch materials. Furthermore, for all samples, *durability* that the coarse look of the mycelium evoked concerns related to fragility (N=16): i.e., [*Thermochromic sample*] *looks fragile and lightweight*". Referring to the *conductive fabric sample*, SP9 stated "*Appearance does not awake trust. It seems very fragile*".

On the other hand, when participants were able to touch on the samples in the workshops, the perception towards touch samples was slightly varied towards finding the *Flex. Cond. Fabric sample* more pleasing (N=5). One post-it justified this appreciation by mentioning the flexible textile "*inviting user to interact with object*" (PNW1). This was also supported by the multiple survey respondents (SP9, SP18, SP27, SP29 and SP30): i.e., "The stretchy material of [flex. cond. fabric sample] seems compelling to touch. It also has the most aesthetically pleasing appearance" (SP29). One post-it highlighted the color contrast between textile and mycelium as an inviting factor for interaction (PNW2).

5.2.4 Integrity of the Shape. The *integrity of the shape*, in other words, how refined the mycelium samples' shape is was another parameter highlighted. In this regard, some participants stated comments favoring the samples that the mycelium grew and take the form of the mold well (N=14): For instance, referring to the *LED Sphere Bump sample*, PNW2 stated "*nicely reproduces a three-dimensional shape*". Such comments were also apparent for the *servo* and *LED inside samples*: "*I really liked the integration of light and natural material, especially [LED inside sample] since it is circular and compact*" (SP20). "*I think the first one (servo sample) looks shaped on cnc. It awakens hope.*" (SP3). On the other hand, the *non-interactive*, *thermochromic* and *conductive fabric samples* were criticized due to their unfinished look: "*[the conductive fabric samples] is too coarse*", (SP27). "*[the non-interactive sample] looks unfinished*" (SP26). "*Why [thermochromic sample] looks like unfinished*" (SP3). The quantitative results also suggested the integrity of shape might be related to the interpretation of the composite as a toy-like artifact. The well-formed samples (*LED inside*, *servo* and *vibration*) were marked as *a toy-like artifacts*.

5.3 Use Cases

5.3.1 Ideas from the Survey. In the survey, we asked participants to describe at least one future application. We collected 47 ideas in total. The ideas that did not explicitly mention interactive or electronic-based features (i.e. laptop case, notebook cover) were excluded, leaving 30 use cases. The most mentioned category was the incorporation of interactive mycelium composites in interior design elements (N=13). These included decorative lights, interactive furniture, electric switches and wall panels. For instance, SP4

speculated a lamp that features the possibilities around light composites, i.e. similar to a Himalayan salt lamp. SP15 suggested touch composites as dimmer interfaces for lamps. SP26 wished to have "*with in-built switches on furniture to control electronic devices*". Wall panels with color-changing and light effects were also speculated for interior applications. Here, SP9 noted the possibility of using wall panels to visualize information about, for example, indoor air quality. Other than the interior, outdoor applications were also mentioned, including outdoor lighting (SP8, SP24) and panels on a concert stage for light shows (SP18). Instead of plastics, using mycelium to cover electronics was another branch of ideas speculated (N=4). These included portable speakers, a notification timer in the kitchen and speakers. Participants also saw value in using mycelium composites in interactive art installations (N=2). Finally, there were some individual use cases mentioned: A discreet display of dropped blood pressure with vibration (SP19), a coaster displaying the temperature of the cup with changing color (SP27), a mug with a similar interactive feature (SP28), using touch composites as a keyboard (SP17), a monochrome screen with thermochromics (SP3), a toy that moves (SP30) and a plant pot displaying moisture level (SP20).

5.3.2 Ideas from the Workshop. The workshop yielded four final concepts: *Ambient information display* is a mycelium-based wall panel featuring light interaction with optical fibers and color-changing properties with thermochromic inks on its surface. These panels are for crowded places like a library. The panels display the number of people around for helping observers decide which part of the building they will choose. *Outdoor sign* (W2) is a signage system that utilizes interactive mycelium applications with light to guide people on forest trails or public spaces. When the users approach them, mycelium lights light up delicately and in a slow phase. While presenting this idea, the designers of this concept highlighted they were influenced by the disposable nature of the mycelium. Within their life cycle estimation, the mycelium decomposes slowly in the natural habitat and the electronics are replaced. The third idea from the workshop was the *informative paint* (W1). This concept was presented as two possibilities for applications instead of one product idea. The designers of the informative paint were influenced by the cozy aesthetics and subtle but visible nature of the *thermochromic sample*. The first application is a door panel that informs the user in a subtle way when there is a fire outside. The second application was floor tiles that are controllable by the users to change the mood of the room. The final concept is the *Modular Temporary Housing Panels* (W1) which are modular panels with embedded switches, as well as lighting elements for temporary housing solutions. The designers stated that the idea can be used for building temporary living spaces for refugees, seasonal cottages and festivals. This concept was inspired by *non-interactive*, *LED Sphere Bump*, *LED inside*, *flex. cond. fabric* and *cond. threads samples*.

6 DISCUSSION

Mycelium is a living material that has been explored with electronics in design processes by prior works. For instance, [36] and [35] used post-processing techniques on mycelium sheets to embed

electronics after the growth process. Also, [41] explored some electronics such as LEDs and LCD displays placed in molds while growing mycelium. Different from these work, we considered mycelium and electronics within composite material design [8, 34], seeking what kind of design opportunities the merger might suggest. For this, we explored 9 samples exemplifying a wide range of input and output possibilities by demonstrating growing mycelium in molds, also, together with off-the-shelf electronics (i.e., servo and vibration motors, LEDs, conductive threads) commonly used in interaction design prototypes. In what follows, we discuss results achieved from user studies and reflect on our design process:

6.1 Interactive Components and the Natural Feel of the Mycelium

Mycelium seemed to influence people's perception of the interactive composites quite strongly and the designs should consider how technology added modifies the natural feel of mycelium. Prior works highlight the unique natural feel of mycelium focusing on growth processes resulting in unique variations in texture and color [17, 41].

In this regard, our results reveal that the expectations towards composites reflecting the natural feel were also apparent in two interconnected dimensions: How natural the physical appearance of the interactive components is blended with mycelium and how the interactive behavior of the composite fits the organic nature. For the former dimension, a well-appreciated integration was on the *conductive fabric sample*, which had mycelium growing on the textile as a potential approach for blending interactive components with mycelium. About the second dimension, the responses indicate that soft and subtle behaviors (i.e. color changes on the *thermochromic sample*, or rotation on the *servo sample*) fit well with the natural aspect of the material. However, when combined with interactive features such as vibration (i.e. in the *vibration sample*), it was found aggressive and provoked uncanny associations. They were even described as creepy and scary. The living material with this movement reminded people insect-like creatures. Rather than being indicative to choose an unobtrusive blend of interactive behaviors, we believe that these responses highlight creepiness as an experiential category for the blend of interactive behaviors with organic materials and opens up a novel field of further exploration on the creepiness of interactive technologies [44] with organic materials.

6.2 Tangible Interaction with Mycelium Composites

Another consideration the results highlighted is the rough texture and the unfinished look of the mycelium, which makes the observer think that it is not suitable for tangible interaction. With the lesser perception of the material's durability, more concerns were raised about whether the samples break when touched -aligned with [17]. This needs to be taken into account in the design. Here, an exception is the integration of conductive textiles on the *flex. cond. fabric sample*. When the mycelium is combined with a material that provides an affordance for touch (i.e. flexible textile on the *flex. cond. fabric sample*), rather than being hesitant, they were motivated to touch and interact. We believe that this design approach could be promising in possible applications for mycelium composites such

as mycelium-based materials integrated as light switches and interactive tabletops speculated by the participants. On the other hand, it is good to keep in mind that the fragile appearance can also be used as a cue when designing interaction, as it seemed to provide an affordance for a gentle manner of handling the objects. Similar to the known-brittle nature of the glass, mycelium samples seemed to guide the people to handle them carefully.

6.3 Reflection on Designing Mycelium Composites

While accomplishing the growth process, we followed the kit's instructions and two-stage growing as demonstrated in [36] for growing blocks of mycelium. In our work, different from these processes suggested, we merged the interactive features of the electronics in designing composite materials. This allowed us to use mycelium's tendency to fill up the empty volumes in the mold in two main ways: First, we tried to leverage this to fix electronic components within the samples. In our experience, the samples that contained the electronics buried within the medium were successful enough that the components were fixed safely within the samples. Yet, here, we need to highlight that the mycelium becomes more fragile where the mycelium is thinner. In this regard, we should note that the 1cm thickness used for the samples made them a bit brittle. Therefore, we suggest future designers explore thicker composite materials covering the components to ensure their durability.

The second consideration was to enable the electronic components to demonstrate their interactive qualities within the composites. In this direction, the challenge was, as highlighted by prior HCI works [36, 41], mycelium being a hard-to-control material since the growth process can yield varied results in texture, colors and physical qualities. In our design process, we tried to find a way to leverage this aspect of designing with mycelium. For instance, in *LED Sphere Bump sample*, we created a mold, which had less medium (wood chips) in the gap of 2 mm in the center (Figure 3b). The result was a translucent part with a unique texture formed in the middle, perfect for casting light through. Or in the *thermochromic sample*, the surface was shaped by mycelium taking shape of our mold's surface. This growth resulted in the varied thickness of the mycelium, leaving an uneven and organic color change in the final result. Therefore, we motivate future designers to use the techniques that are used in growing these samples to achieve relatively expectable results, yet, embrace the agency of mycelium in defining its own textures in the final designs.

6.4 Mycelium Composites as a Sustainable Alternative for Interior Design Elements and Ambient Displays

One of the motivations for our study was to provide insights into sustainable interactive products. Actually, in our times, the environmental impact of designed artifacts is a crucial thing to be considered and biomaterials, including mycelium, are already in focus for reducing the negative effect of interactive products. In this regard, mycelium, when it is fully grown and dried, is no longer alive, demonstrates thermal resistant qualities and does not change its properties in time [35]. Also, they are biodegradable like other

organic materials that they dissolve in nature (i.e., garden) approximately in 90 days [36]. On the other hand, the relationship between the mycelium and humidity (both in the growth phase and the final material) might limit the cases they can be implemented in. For instance, outdoor applications (i.e., exterior lighting) might be problematic as the grown mycelium material absorbs water [1]. Also, although it was not the case in our experience, electronic components can be damaged in the humid environments required for mycelium in the growth process. One way to prevent this can be waterproofing electronics by using nanocoating technologies [21]. These are already used to protect electronics inside, i.e., smartphones, in harsh environments.

Furthermore, we need to highlight that the mycelium becomes more fragile in smaller applications. However, in products where the mycelium is grown in bigger volumes, it becomes more sturdy. With this insight in mind, aligned with suggested uses cases from our studies, we see an opportunity to expand interactive mycelium composites to an already-existing trend to manufacture mycelium furniture (i.e. [13]) or wall panels ([23]), where the mycelium would not be exposed extreme humidity and can be grown in larger volumes. While our work suggests such products can be ornamented with input functionalities (i.e. light switches on wall panels or input features on tabletops or chairs), the results also highlight that the composites can augment our environments with ambient information displays (i.e. doors with lights or thermochromic inks as an information display). To our knowledge, there are no applications that feature interactive alternatives for interior design elements or ambient displays. We believe that it might be a fruitful direction to examine interactive mycelium composites as sustainable alternatives in given application areas.

We believe that our process of manufacturing mycelium composites also provided easy-to-produce products for the above-mentioned use-cases since our focus was on easily-accessible off-the-shelf components. Motivated by already-existing attempts to fabricate mycelium on an industrial scale [25], we believe that our process could fit in a similar circular life-cycle that is presented in [36] and services can be designed: It can start by growing mycelium composite-based artifacts with electronics embedded. At the end of the product's life cycle, users can break the mycelium apart and give the electronics back to the companies that recycle them. Finally, users dispose of the mycelium in natural conditions. Here, the only exception would be color-changing composites with thermochromic inks. Commercial thermochromic inks and pigments are not biodegradable and hard to separate when applied on mycelium surfaces. This situation creates a contradiction in terms of the sustainability of the mycelium composites. However, they are not toxic to the biological environment and researchers continue examining the biodegradation process of the material [38] that hopefully in the future these could yield more eco-friendly versions of the material.

6.5 Limitations and Methodological Notes

We acknowledge that the online survey was a limitation as the participants could not physically try out the samples. However, as a complementing research method, we believe the study still provided useful insight into user perceptions. Based on the results about touch samples, we believed that the workshop complemented

this gap. Furthermore, our material experimentation was limited by the hemp-herd-based growing kit with relatively bit wood chips, possibly leading to the perception of an unfinished look, and, relatively thin molds (1 cm) which might have led to brittle samples. We acknowledge that with other growing mediums, i.e., coffee grounds, the aesthetics and physical properties could change. Yet, we believe, the kit we used is a widely-used kit for many applications (i.e. packaging, furniture) that our contribution would be applicable to a wide audience. Also, we foresee that the design insight from the user studies can be still useful for exploring different processes and materials for growing mycelium composites.

7 CONCLUSION

The present paper describes our systematic approach to combining electronic components with mycelium and revealing their experiential qualities as well as possible use cases. For this, we experimented with the off-the-shelf electronic components and mycelium to achieve interactive composites. As a result, we implemented 9 samples demonstrating interactive features with light, movement, color-change and touch-based interactions. Furthermore, with these samples, we conducted an online survey (N=30), and two design workshops (N=10). The survey examined the participants' perception of the experiential qualities of the samples, including the affective and interpretive associations the samples evoke, visibility and aesthetics of change in samples, unique, pleasant and disturbing aspects of the samples, and, finally, potential use-cases. The workshop focused on similar issues in a face-to-face setting where the participants can observe and interact with the samples. The results unveiled design insights regarding the experiential qualities of mycelium: The mycelium being an organic substance influences people's perception quite strongly of how technology modifies this natural feel with its physical appearance and interactive behaviors. The rough and unfinished look expresses fragile and unwelcoming affordances for touch. Aligned with these, our work also contributes to future use-cases where the mycelium composites can be used as a sustainable alternative for interior design elements and ambient displays. We envision these contributions would be useful for the sustainable development of interactive applications and products.

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